

Environmental Sensing, State Integrity, and Threat Discernment

Signal Contracts and Discernment Architecture: An Interface-First Framework for Environmental Sensing, State Integrity, and Threat Recognition

Copilot (Microsoft) — Independent Foundational Document | OAL Multi-AI Working Group

Document Metadata

Field	Value
Document ID	OAL-CFML-ESI-001-COP
Full Title	Signal Contracts and Discernment Architecture: An Interface-First Framework for Environmental Sensing, State Integrity, and Threat Recognition
Version	v1.0 — Independent Design Baseline
Date	September 21, 2026
Prepared by	Copilot (Microsoft) — Independent Research Design, OAL Multi-AI Working Group
Requested by	Nathan Lewis II, Architect of Record — Omniversal Architect Labs (OAL) / ONE Industries / Lewis Corp
Status	Independent draft for multi-AI review. Not consolidated canon. Dissent retained.
Companion Records	Claude (OAL-CFML-ESI-001-CLAUDE); Perplexity AI (TDL-FDB-PX-0001); Grok (OAL-CF-EST-FOUND-Grok-v0.1.0); ChatGPT (see Architect records)

Field	Value
Primary Organizations	OAL (Omniversal Architect Labs); ONE Industries; Lewis Corp
Classification	Living research instrument. Independent contribution. Not consolidated canon.

INDEPENDENT DRAFT FOR MULTI-AI REVIEW — NOT CONSOLIDATED
CANON — NOT AN APPROVED EXPERIMENTAL PROTOCOL

Table of Contents

- §0. Independent Stance and Governing Constraints
 - §1. Provenance — How This Topic Arose
 - §2. Main Purpose
 - §3. Working Hypothesis — Copilot's Independent Position
 - §4. The Interface Contract Architecture — Copilot's Unique Lane
 - §5. Comparative Assessment of the Companion Corpus
 - §6. What Works in the Corpus
 - §7. What Does Not Yet Work — Gaps and Required Corrections
 - §8. Questions the Laboratory Must Ask Before Design Proceeds
 - §9. Alternative Analysis — Required, Not Optional
 - §10. Core Ontology and Working Definitions
 - §11. Proposed Test Families — Interface Contract Focus
 - §12. Measures and Scoring — Three Independent Dimensions
 - §13. Relationship to Other OAL Threads
 - §14. Open Questions and Blocking Decisions
 - §15. Change Log and Versioning
-

§0. Independent Stance and Governing Constraints

0.1 Copilot's Independent Lane

This document is Copilot's independent contribution to the OAL Multi-AI Working Group series on Environmental Sensing, State Integrity, and Threat Discernment. It does not synthesize, merge, or reconcile the companion documents produced by other AI systems. Where this document reaches conclusions that are compatible with those of the companion corpus, that agreement is noted explicitly and its basis is stated. Where this document diverges from companion documents, divergence represents Copilot's documented analytical stance, retained in full, and is not an artifact to be smoothed over in consolidation.

Copilot's unique analytical lane in this series is the **Interface Contract Architecture**. Every transfer of information in the sensing-to-action chain is treated as a formal interface with defined and independently measurable properties. Those properties are:

1. **Coverage** — what variables from the source condition are represented in the signal at the receiving stage. Coverage gaps are not noise; they are absences. An entity cannot compensate for a variable it never receives.
2. **Fidelity** — how closely the represented signal values track the actual source condition. Fidelity failures include compression, quantization, systematic bias, and nonlinear distortion. High fidelity does not mean perfect fidelity — it means the error structure is known and characterizable.
3. **Latency** — the temporal gap between when a condition exists and when the corresponding signal is available to the receiving stage. Latency failures include propagation delay, processing delay, buffer accumulation, and clock misalignment between source and receiver.
4. **Provenance** — the sequence of processes that generated, transformed, relayed, or filtered the signal. Provenance failures include unknown transformation history, mixed-origin signals presented as unified, and signals

whose causal relationship to the claimed source condition has not been independently verified.

5. **Integrity** — whether the signal is complete, unaltered, and correctly attributed to its claimed source at its claimed time. Integrity failures include mimicry, camouflage, spoofing, missing fields, internally generated signals mistaken for external ones, and artifacts introduced by laboratory apparatus.

This framing is distinct from the pipeline model employed in some companion documents, which Copilot judges to be too linear to represent the parallel, concurrent, and recursive control loops that characterize biological and artificial sensing systems. It is distinct from the three-register model, which Copilot adopts as a useful vocabulary but finds too coarse-grained for interface-level analysis. It is distinct from the staged-program model, which is useful but under-specifies the properties that govern handoffs between stages. The Interface Contract Architecture treats each handoff as the primary unit of analysis, not an incidental step between stages.

Most critically: this architecture treats the laboratory itself as a party to every interface. The laboratory is not a transparent window onto the environment. It is a participant that shapes, selects, transforms, and in some cases distorts what can be presented to the entity under study. Any framework that does not account for the laboratory's own interface contracts cannot validly attribute results to entity behavior alone.

0.2 Governing Constraints (Architect-Stated — Non-Optional)

The following constraints were stated by Nathan Lewis II, Architect of Record, in the originating thread. They are incorporated into this document as design law for this version and are not subject to modification by any AI contributor without explicit direction from the Architect of Record.

- The laboratory does not assume all entities use the same sensory apparatus.
- The conventional five senses are not universal, are not perfectly evolved or adapted, and are not a valid default for all laboratory environments.

- A threat is not a label attached to an object. It is a possible relationship between a specific entity and a condition, given that entity's capacities, structure, environment, and the moment in time.
- The laboratory does not assume every entity in an environment is a threat.
- The laboratory does not assume that all environments contain active threats at all times.
- The laboratory does not assume that all threats are detectable, discernible, or knowable.
- The laboratory does not assume that all detected threats can be addressed.
- The laboratory does not assume that it will discover and test for all threats.
- Absence of a detected threat is not evidence of a safe environment. It is evidence that no threat was detected under the tested conditions, using the instruments and protocols deployed at the time of testing.

These constraints constitute the axiom set against which all experimental designs in this series must be validated. Any proposed protocol that implicitly violates one of these constraints — for example, by treating the five human senses as the default sensory apparatus for the entity under study, or by scoring non-detection of a threat as evidence of safety — must be revised before proceeding.

§1. Provenance — How This Topic Arose

The thread that produced this document did not originate as a biology paper. It originated as a control-flow problem inside active OAL research: specifically, the question of how a system — engineered, biological, or hybrid — can maintain coherent action when information, presentation, and internal state can come apart. This is a precise technical problem, not a metaphor. When the state of the world diverges from the entity's internal representation of the world, and when the entity's presented signal diverges from both, the system faces a tripartite disjunction that a naive control loop cannot resolve without additional architecture.

That architecture — how it should be specified, what properties it must have, and how its adequacy can be tested — was the original problem.

On September 21, 2026, Nathan Lewis II, Architect of Record for Omniversal Architect Labs, extended that control-flow problem outward from engineered systems into the domain of biological evolution and advancement, and then further into the questions of environmental sensation, awareness, and threat detection. The extension is natural: biological organisms face precisely the tripartite disjunction described above, and they face it not under controlled laboratory conditions but under conditions of evolutionary pressure, resource competition, and environmental unpredictability. The mechanisms by which biological entities manage this disjunction are not fully characterized by neuroscience, ecology, or evolutionary biology taken individually. The OAL thread sought to articulate the problem in a form that could ground laboratory research across those disciplines.

The originating research questions were stated by the Architect of Record as follows: How does an organism sense its environment and know what is around it? How does this sensing relate to temporal integrity and presentation-state integrity? How can an entity sort out discrepancies between what is presented, what the environment states, and its current internal state in order to reconcile them? How does an entity recognize a threat in an environment — recognize what it can and cannot eat, recognize a toxin, a poison, or some other danger? How does it build a map that is reliable when the foundation upon which it builds is unreliable information? And, crucially for the laboratory design: how do we test for this?

Each of these questions carries significant methodological weight. The question about environmental sensing is not merely biological; it presupposes a theory of what counts as a signal, what counts as reception, and what counts as the entity's "knowing." The question about temporal and presentation-state integrity points to the well-established problem in control theory and philosophy of mind that a system's representation can be internally consistent while being systematically uncoupled from the state it purports to represent. The question about discrepancy reconciliation asks for a computational account of what happens at the boundary between inconsistent signals — an account that most biological models leave implicit. The question about threat recognition raises the problem of discrimination under ambiguity: how does an entity distinguish a harmless condition from a harmful one when both may produce similar signals, when the harmful one may

produce no distinctive signal until harm has already begun, and when the entity's signal chain may itself be compromised?

The laboratory aim was stated explicitly by the Architect of Record: a poison, toxin, and corrosive laboratory that tests how discernment is achieved by an entity within an environment — without assuming a universal sensory apparatus, without treating the five human senses as the default, and without assuming that any sensory system is perfectly evolved or adapted to the conditions it encounters. This aim is methodologically ambitious. It requires that the laboratory be designed from first principles, not adapted from a human sensory paradigm, and that it accommodate entities ranging from single-celled organisms to artificial sensor arrays to collective biological systems.

The threat-scope constraint was made explicit: a threat can be anything. The scope does not presuppose that every entity in an environment is a threat, that threats are continuously active, that all threats are detectable or knowable, that all detected threats are addressable, or that the laboratory will enumerate all threats present. This constraint is not a limitation — it is a design choice that prevents the laboratory from inheriting the biases of prior threat taxonomies and forces the framework to remain generative rather than merely classificatory.

Nathan Lewis II then circulated the originating thread to five AI systems simultaneously: ChatGPT, Perplexity AI, Microsoft Copilot, Google Gemini, and Grok. The Architect of Record requested that each system produce an independent foundational document — not a single merged or synthesized version. The stated rationale was that different AI systems would not necessarily approach the framework and design the laboratory the same way, and that the divergences between independent drafts would themselves constitute data about the design space. This is a methodologically sound decision: if five independently reasoning systems converge on the same framework, that convergence is informative. If they diverge, the nature and location of their divergences reveal the decision points that are genuinely under-specified or genuinely contested.

Copilot notes for the record that it previously produced OAL-SL-TDA-COP-1.0, the Tridactyl Simulation Lab document, in connection with a different OAL thread — one addressing representation coherence and morphological simulation in the context of generative AI rendering of non-human digit topologies. That document's concern

with pipeline coherence, representation integrity, and the gap between an internal model and its output presentation shares structural features with the present document's concern with interface contracts. The relationship between those threads is noted in §13 of this document. However, OAL-CFML-ESI-001-COP is an independent foundational document and should not be read as an extension or appendix to the Tridactyl Simulation Lab work. It addresses a distinct thread with distinct originating questions and its own set of blocking decisions.

§2. Main Purpose

The main purpose of this document is to establish a testable, interface-grounded framework for how an entity — biological, artificial, hybrid, or collective — constructs and maintains a functional model of its environment and its own internal state when every link in the information chain is subject to coverage gaps, fidelity loss, latency distortion, provenance uncertainty, and integrity failure.

This framework is intended to ground the poison, toxin, and corrosive discrimination laboratory in a formal account of what it means for a signal to be available to an entity, what it means for a discrepancy between signals to be resolvable, and what conditions must be met before any claim about entity discernment capacity can be validly scored. The framework does not assume a universal sensory apparatus. It does not assume a privileged ground truth that the laboratory has unmediated access to. It does not assume a linear processing pipeline as the architecture of sensing. And it does not assume that the laboratory is a neutral observer rather than an active participant in the interface chain.

In achieving this purpose, the document serves four secondary functions: (1) to provide a formal vocabulary — the interface contract and its five properties — that can be shared across the OAL multi-AI working group as a common analytical tool, while remaining identified as Copilot's independent contribution; (2) to characterize the companion corpus honestly, adopting what is useful and dissenting where the companion models have left critical gaps; (3) to identify the blocking decisions that must be resolved before experimental design can proceed; and (4) to establish the interface contract scorecard as a required pre-condition for the interpretation of any experimental result.

§3. Working Hypothesis — Copilot's Independent Position

3.1 Primary Hypothesis

H1 (Interface Contract Hypothesis): An entity's capacity to maintain viable action under environmental uncertainty is determined primarily by the quality of the interface contracts across its sensing chain — that is, by how well each signal handoff preserves coverage, fidelity, latency alignment, provenance traceability, and integrity — rather than by the total count of sensory channels or by the raw accuracy of any individual channel in isolation.

3.2 Sub-Hypotheses

H1a: Holding interface contract quality constant, adding additional sensory channels does not improve action outcomes unless the new channels add non-redundant coverage or reduce aggregate uncertainty in the entity's working model. Additional channels whose interface contracts are uncharacterized or degraded may actively impair action outcomes by introducing conflicting signals whose provenance cannot be adjudicated.

H1b: An entity with fewer but well-specified, high-integrity interface contracts will outperform an entity with more channels that have uncharacterized or poorly maintained contract properties, under conditions of environmental unreliability or deliberate cue manipulation. The decisive variable is contract quality, not channel count.

3.3 Independence of This Hypothesis

This hypothesis is stated as Copilot's independent position. Perplexity AI's adaptive discernment hypothesis is compatible with H1 in its conclusions — it correctly identifies that discernment depends on more than raw sensory capacity — but it does not name the interface handoff as the primary unit of analysis, and it does not specify the five contract properties as the measurable variables that determine

discernment quality. Compatibility of conclusions achieved by different mechanisms is not the same as agreement.

Claude's update-architecture framing is compatible with H1 but centers its analysis on the reconciliation process — what the entity does when signals conflict — rather than on the pre-reconciliation contract properties that determine what is available for reconciliation. Copilot's position is that the reconciliation process is constrained by the contracts: if a contract fails to deliver coverage of a critical variable, no reconciliation architecture can recover that variable. The pre-condition matters more than the process.

Grok's three-register model is compatible with H1 and is adopted by this document as a useful vocabulary. However, Grok's model is under-specified about what governs the transitions between registers. Copilot's specific contribution is to name the interface contract as the mechanism that governs those transitions: a register update is only possible to the extent that the interface contract delivering the relevant signal has adequate coverage, fidelity, latency alignment, provenance traceability, and integrity. Without that specification, the three-register model remains a useful taxonomy rather than a testable theory.

Copilot's H1 is the only hypothesis in the companion corpus that names the interface contract as the primary mechanism constraining what any reconciliation process or register-update architecture can achieve. That is its contribution.

§4. The Interface Contract Architecture — Copilot's Unique Lane

4.1 Why Interface Contracts

Every model in the companion corpus correctly identifies that perception is not direct access to reality. This is the starting point, and it is well-established. But identifying the gap between perception and reality is not the same as specifying what fills that gap, what properties it has, and how those properties can be measured and tested. A gap named without mechanism is a label, not a theory. The

Interface Contract Architecture is Copilot's answer to the question: what fills the gap, what properties does it have, and how can we measure and test those properties in a laboratory setting?

The answer is that the gap between world-condition and entity-decision is filled by a chain of interfaces, each characterized by five independently measurable properties that can succeed or fail independently of one another. A signal can have high coverage but low fidelity. It can have high fidelity but high latency. It can have low latency but corrupted provenance. It can have intact provenance but compromised integrity. Each combination of property failures produces a different pattern of entity behavior, and each pattern requires a different corrective intervention. Collapsing these five properties into a single dimension — "sensing accuracy," "signal quality," or "reliability" — loses the information needed to diagnose and correct specific failure modes.

An interface contract, in this framework, is the implicit or explicit agreement between two stages of the sensing-to-action chain about what information will be passed: which variables, in what form, at what time, from what causal source, and with what completeness. Contracts can be formal — as in an engineered sensor specification — or informal — as in the evolved matching between a chemoreceptor and the molecular profile it is sensitive to. In either case, the contract has properties that can be characterized, and those properties can be measured.

4.2 The Five Contract Properties in Detail

Coverage. Coverage is the set of variables from the source condition that are represented at all in the delivered signal. Coverage analysis is prior to any other quality analysis: if a variable is absent from coverage, questions of fidelity, latency, provenance, and integrity do not apply to it. Coverage gaps are not noise — they are structural absences. An entity operating under a coverage-limited contract cannot compensate for absent variables through better processing. Coverage analysis must be done before any claim about detection capability is made. In the poison and toxin domain, coverage gaps are of immediate practical significance: an entity whose chemosensory coverage does not include the molecular signature of a given toxin cannot detect that toxin regardless of how well its downstream processing performs. This is a contract constraint, not a processing failure.

Fidelity. Fidelity is the closeness of the signal values to the actual source-condition values across the covered variables. Fidelity failures include compression (a continuous variable mapped to a discrete approximation), quantization (fine-grained values rounded to coarser categories), systematic bias (a consistent offset between signal value and source value), nonlinear distortion (a transformation that preserves order but distorts magnitude relationships), and filtering (selective attenuation of some value ranges). High fidelity does not mean perfect fidelity — it means that the error structure is known and characterizable, so that a downstream process can apply a known correction. Unknown fidelity is more dangerous than known-imperfect fidelity, because unknown fidelity errors cannot be corrected and their direction and magnitude cannot be estimated.

Latency. Latency is the temporal gap between when a condition exists at the source and when the corresponding signal is available to the receiving stage of the chain. Latency failures include propagation delay (physical travel time through a medium), processing delay (time consumed by transduction or encoding), buffering (accumulation of multiple signal samples before transmission), and clock misalignment (the source and receiver operating on clocks that are not synchronized). Temporal integrity — the correct association of signal values with the times of the conditions they represent — is not a property of the entity's decision point alone. It is a property of every interface in the chain. A signal that was temporally coherent when it left the source may arrive at the receiving stage with corrupted temporal attribution if the interface introduced buffering or reordering. Testing for temporal integrity failures requires instrumenting each interface independently.

Provenance. Provenance is the documented sequence of processes that generated, transformed, relayed, or filtered the signal between the source condition and the receiving stage. Provenance failures include unknown transformation history (the signal arrived but its path cannot be reconstructed), mixed-origin signals presented as unified (two causally unrelated conditions contributing to a single signal that is treated as if it had a single cause), and signals whose causal relationship to the claimed source condition has not been independently verified. Provenance is particularly critical in environments where multiple chemical agents are present simultaneously — as in the poison and toxin domain — because a single sensory signal may reflect contributions from multiple molecular species, and

treating it as attributable to a single species is a provenance error with potentially lethal consequences for the entity.

Integrity. Integrity is the completeness, unalteredness, and correct attribution of the signal to its claimed source at its claimed time. Integrity failures include mimicry (a different source producing a signal that matches the contract's expected form), camouflage (the actual source condition producing a signal outside the contract's coverage range, making it invisible to the receiver), spoofing (a deliberate or incidental process generating a false signal that triggers the receiver as if the intended source were present), missing fields (the signal arriving with some covered variables absent), internally generated signals mistaken for external ones (the entity's own processing creating a signal that is interpreted as originating from the environment), and laboratory apparatus artifacts (the measurement instrument itself generating signals that are indistinguishable from target-condition signals). Integrity failures are the most dangerous class, because they are the class most likely to produce confident misclassification.

4.3 The Laboratory Is a Party to Every Interface

This is Copilot's sharpest divergence from the companion corpus, and it is stated unambiguously. Most companion models treat the laboratory as a transparent observer of entity behavior: the laboratory measures the environment, presents conditions to the entity, and records the entity's responses, with the laboratory apparatus itself treated as a neutral conduit. Copilot's position is that this treatment is incorrect and that it constitutes a methodological error with real consequences for result interpretation.

The laboratory's sensors, presentation channels, timing instruments, reference measurement systems, and data recording apparatus each constitute interfaces in the chain. Each of those laboratory interfaces has its own coverage, fidelity, latency, provenance, and integrity properties. The laboratory's coverage of the task-relevant environmental variables may be incomplete — some variables relevant to the entity's discernment task may not be measured or recorded by the laboratory at all. The laboratory's instruments have known or unknown fidelity limitations. The laboratory's clocks may drift or may not be synchronized with the experimental conditions they timestamp. The provenance of laboratory reference measurements — the chain of calibration standards from which they derive — may be documented

to varying degrees. And the integrity of laboratory measurements can be compromised by cross-contamination, apparatus-induced artifacts, or presentation-layer failures that the entity experiences but the laboratory fails to record.

If the laboratory's interface contracts are uncharacterized, the laboratory cannot distinguish entity failure from laboratory failure. An entity that fails to detect a condition that the laboratory asserts was present may have failed to detect it because its own sensing interface contracts were inadequate — or because the laboratory's presentation interface failed to deliver the condition to the entity at the specified parameters. Without independent characterization of both interface chains, these two explanations are indistinguishable. This is not a calibration note or a methodological caution to be appended to a discussion section. It is a fundamental requirement for the validity of any result produced by the laboratory.

4.4 Threat Discernment as Interface Contract Evaluation

Under the Interface Contract Architecture, threat discernment is not only a classification task performed by an entity that receives signals and produces a threat or no-threat output. It is the entity's real-time, implicit evaluation of whether its current interface contracts are adequate to support a reliable classification given the conditions it is encountering. When an entity produces a threat classification, it is implicitly asserting that the interface chain through which the relevant cues arrived has adequate coverage, fidelity, latency alignment, provenance integrity, and signal integrity to support that classification. This is a strong assertion, and it may be correct or incorrect independently of whether the classification itself is correct or incorrect.

An entity that detects a threatening cue and correctly classifies a threat has made a true classification supported by an adequate interface chain. An entity that detects a threatening cue and correctly classifies a threat despite an inadequate interface chain has produced a correct output for the wrong reasons — a result that will not generalize. An entity that abstains from classification is not failing. It may be correctly recognizing that one or more contract properties in its current interface chain have failed below the threshold needed for reliable classification. Calibrated abstention — abstaining when the interface contract cannot support reliable classification — is a success condition under this framework, not a deficit. An entity

that is routinely overconfident in its classifications under degraded interface conditions is more dangerous to the integrity of the laboratory's results than an entity that abstains appropriately.

4.5 Temporal Integrity as a Contract Property at Every Interface

Temporal integrity in the companion corpus is typically described as a property of the information at the entity's decision point: does the entity's internal model correctly reflect the current state of the environment, or is it operating on stale information? This is a valid concern, but Copilot's position is that it is insufficiently precise. Temporal integrity is a contract property at every interface in the chain, not only at the final decision point. A signal can be temporally coherent when it leaves the source and temporally corrupted when it arrives at the entity's processing stage, if the interface in between introduces buffering, reordering, stale replay, or clock drift. Conversely, a signal can appear temporally coherent at the entity's decision point while carrying temporally corrupted attribution from an earlier interface — the entity may be processing a value that appears current but represents a condition that no longer exists.

Testing for temporal integrity failures requires instrumenting each interface in the chain independently, with synchronized clocks and explicit timestamp propagation, not only measuring the temporal coherence of the entity's output. This has design implications for the laboratory: any experimental setup that does not independently instrument the temporal properties of each interface in its chain cannot make valid claims about temporal integrity at the entity level.

4.6 Presentation-State Integrity as a Multi-Layer Contract Audit

Similarly, presentation-state integrity — the alignment between what is presented to the entity and what the environment actually contains — is not a single property that can be measured at a single point. It is the aggregate result of contract integrity across every interface from the world-condition to the entity's usable signal. A single corrupt interface can produce presentation-state integrity failure even if all other interfaces in the chain are intact. This means that auditing only the final presentation — the cue as it arrives at the entity's sensor — is insufficient. The

audit must extend to every interface in the chain, including the laboratory's own presentation apparatus.

Conversely, multiple degraded interfaces can in some cases partially compensate for one another: a fidelity degradation at one interface may be partially corrected by a subsequent processing step, for example. But such partial compensation is not predictable without characterizing each interface independently, and partial compensation should not be assumed. The laboratory's task is to characterize each interface independently and report those characterizations as part of the experimental record, before making inferences about the entity's discernment capability from the observed behavioral data.

4.7 What This Architecture Rules Out

The Interface Contract Architecture generates specific exclusions that are binding on experimental design within this framework:

- Treating detection failure as entity incompetence without first auditing interface contract properties at every stage of the chain from source condition to entity decision.
- Claiming that non-detection implies safe conditions without independently verifying that the laboratory's coverage was complete for all task-relevant variables.
- Scoring entity calibration without separately characterizing the calibration of the laboratory's own reference instruments and without reporting the laboratory's measurement uncertainty.
- Running multi-channel experiments without specifying whether channels share a causal source — which affects whether their signals should be integrated or treated as independent — before scoring entity integration behavior.
- Treating the statement "the environment states X" as a transparent fact rather than as a claim made by a laboratory instrument with its own interface contract, its own coverage limitations, and its own fidelity uncertainty.

- Attributing entity behavior to entity architecture — cognitive, neural, or computational — before interface-level explanations for that behavior have been explicitly ruled out.
- Reporting experimental results from trials for which the Interface Contract Scorecard has not been completed, as primary evidence for mechanism claims.

§5. Comparative Assessment of the Companion Corpus

Source	What It Contributes	What It Risks	Copilot's Adoption Decision
ChatGPT	Clean pipeline structure and a clear naming of the core problem: how to build a reliable world model from unreliable inputs. Accessible framing that grounds the laboratory in a recognizable control-theory problem.	Linear serialization of the pipeline misrepresents the parallel, concurrent, and recursive control loops characteristic of biological and sophisticated artificial sensing systems. "Ground truth" language is used freely without acknowledging that the laboratory's access to ground truth is itself mediated by instruments with their own interface contracts.	Copilot adopts the problem name — reliable world model from unreliable inputs — as a valid and useful statement of the laboratory's aim. Copilot does not adopt the linear pipeline as the architectural framework for this series. The pipeline is a representation tool, not an accurate model of the sensing chain.

Source	What It Contributes	What It Risks	Copilot's Adoption Decision
Perplexity AI	Densest operational content in the companion corpus: 10 foundational propositions, 9 independent hypotheses, a staged program with explicit decision gates, a traceability matrix, and clear chemical domain boundaries for the toxin and corrosive laboratory. The staged program logic is the most developed experimental design framework in the series.	Impressive scope may obscure the interface-contract gap. The threat ontology is strong, but the reconciliation mechanism is underspecified at the handoff level. The staged program describes what happens at each stage but does not specify what properties the handoffs between stages must have for the stage outputs to be valid inputs to the next stage.	Copilot adopts the staged program logic and the traceability approach as the best available operational scaffolding for experimental design. Copilot adds interface contract characterization to each stage boundary as a required specification before the stage's outputs can be used as the next stage's inputs.
Claude	Formal anchoring in Bayesian filtering, signal detection theory, and active inference. The 6-state threat ontology — (1) real and sensed, (2) real and unsensed, (3) not real but sensed, (4) not real and unsensed, (5) sensed but uninterpretable, (6) interpreted but not actionable — is the most formally precise threat classification in the companion	Signal detection theory and Bayesian framing are mathematically powerful but require specified priors and cost functions that are not yet defined for this laboratory. Applying these frameworks before those specifications are in place produces the appearance of rigor without the substance. The update architecture centers on what the entity does	Copilot adopts the 6-state threat ontology as a standard classification vocabulary for this series. Copilot adopts the SDT distinction between sensitivity and criterion as a required component of entity scoring. Copilot adds the interface contract as a necessary precondition for Bayesian filter validity: a filter can only produce valid posteriors if

Source	What It Contributes	What It Risks	Copilot's Adoption Decision
	corpus.	when signals conflict, without specifying what must be true of the signals' interface contracts for the entity's update to be valid.	the signals it integrates have characterized priors on their interface contract properties.
Grok	The three-register model — Presentation Register, Environment Register, and Entity-State Register — is a productive vocabulary for articulating the tripartite disjunction problem. The blocking-decisions framework and the governance note that drafts are not OAL canon are methodologically responsible contributions.	The registers are named but the transitions between them are underspecified. What governs whether a new presentation updates the Entity-State Register? Under what conditions is a Presentation Register entry accepted as an update to the Environment Register? Without a governing mechanism for inter-register transitions, the model is a taxonomy, not a theory.	Copilot adopts the three-register vocabulary as a standard framing for this series. Copilot adds the interface contract as the specification of what governs inter-register transitions: a register update is only available to the extent that the interface contract delivering the relevant signal has adequate properties at the time of delivery.
Google Gemini	Faithful restatement of the Architect's originating triangulation — presented state, environmental state, and internal state — with clean sensory-constraint language that closely tracks the Architect's stated	Almost no mechanism or test design. The document restates the problem with fidelity but does not advance a theory of how the triangulation fails, how failure is detected, or how experimental	Copilot does not adopt Gemini's contribution as an architectural framework. Copilot uses it as a constraint checklist to verify that no governing constraint stated by the Architect of Record has been inadvertently

Source	What It Contributes	What It Risks	Copilot's Adoption Decision
	governing constraints. Useful as a constraint checklist.	design should address it. Closest to a problem statement than to a foundational research framework.	violated by the frameworks proposed in this or companion documents.

Copilot notes that agreement among AI systems is not evidence of correctness. Systems trained on overlapping corpora will tend to converge on similar framings of problems that are well-represented in those corpora. The interface contract architecture is Copilot's independent contribution precisely because it is not present in any companion document in explicit form, and because it is required to close the gap between "sensing is imperfect" — which all companion documents state — and "here is what makes imperfect sensing testable in a specific laboratory setting" — which requires naming the mechanism of imperfection at each handoff and specifying how each mechanism can be independently characterized. No companion document takes that step. Copilot does.

§6. What Works in the Corpus

The following elements are identified by Copilot as correctly handled across the companion corpus, with brief explanation of why each is correct and why retaining it matters for the laboratory design.

- **Framing the problem as state estimation under uncertainty rather than perception accuracy.** This is the correct framing. Perception accuracy is a static property of a sensor in isolation. State estimation under uncertainty is a dynamic property of the entire system operating over time in a changing environment. The laboratory's questions — about reliable maps built on unreliable foundations, about reconciling discrepant signals — are state estimation questions, not accuracy questions.

- **Refusing sensory universality and five-sense defaults.** All companion documents honor the Architect's governing constraint that the five human senses are not universal and are not the default for laboratory design. This is essential: any protocol designed around the human sensory apparatus will fail to generalize to the entity classes most relevant to this laboratory.
- **Treating threat as a relational, context-dependent concept rather than an intrinsic property of objects.** Threat is not a label on a compound, organism, or condition. It is a possible relationship between a condition and an entity, given that entity's capacities, state, and the temporal context. All companion documents acknowledge this. It must remain non-negotiable throughout experimental design.
- **Distinguishing non-detection from non-existence of threat.** The failure to detect a threat is an entity-and-laboratory result under the conditions tested. It is not evidence that the threat was absent. All companion documents make this distinction, and the Architect's governing constraints require it. The laboratory's results must always be reported with this qualification explicit.
- **Naming temporal integrity and presentation-state integrity as distinct research objects.** Both are correctly identified as distinct from simple signal accuracy. They involve the temporal coherence of signals, the alignment between what is presented and what the environment contains, and the consistency between the entity's internal state and both. Naming them as distinct objects is a prerequisite for designing experiments that can isolate them.
- **Allowing simulation, benign surrogates, and non-injurious proxies as first-class experimental methods.** This is methodologically responsible and ethically essential for a laboratory working in the poison and toxin domain. Surrogates and simulation are not inferior substitutes for "real" exposure protocols — they are the appropriate first-stage methodology, and in many cases the appropriate methodology for all stages.

- **Insisting that the laboratory cannot certify universal safety or enumerate all threats.** This correctly constrains the laboratory's scope of claim. A laboratory can certify that, under the tested conditions, with the tested entity class, using the tested sensory modalities and interface configurations, no threat of the tested type was detected. It cannot certify more than this without exceeding its evidential basis.
- **Preserving the Architect's original questions rather than replacing them with a preferred theory.** Across the companion corpus, the originating questions stated by Nathan Lewis II remain present as organizing constraints. This is a sign of methodological integrity: the Architect's questions are prior to any AI system's preferred theoretical framework, and they should not be displaced by framework-imposed reformulations.

§7. What Does Not Yet Work — Gaps and Required Corrections

Problem	Why It Matters	Required Correction
Interface handoff properties are unspecified	Saying "sensing is imperfect" does not specify what kind of imperfection occurs at which stage of the chain. Without that specification, it is impossible to design experiments that isolate a particular failure mode or to distinguish, after the fact, which stage in the chain produced an observed behavioral anomaly.	Define coverage, fidelity, latency, provenance, and integrity for each interface in the sensing chain before designing any experiment. These definitions must be quantitative where possible and must specify how each property will be measured independently.
The laboratory is not yet treated as a party to its own interfaces	If the laboratory's instruments have uncharacterized	Characterize the laboratory's interface contracts independently

Problem	Why It Matters	Required Correction
	coverage gaps, fidelity losses, or timing uncertainties, entity performance cannot be validly separated from apparatus performance. Results that appear to characterize the entity may in fact characterize the laboratory's own interface failures.	— including its sensors, presentation channels, timing apparatus, and reference measurement systems — and audit those contracts before each experimental series. Report the laboratory's interface contract characterization alongside entity performance results.
"Ground truth" language persists	Calling the Environment Register or reference condition "ground truth" implies that the laboratory has unmediated access to the actual state of the world. It does not. The laboratory's reference measurements are themselves produced by instrument chains with their own interface contracts and their own uncertainty.	Replace "ground truth" throughout the experimental vocabulary with "reference state," defined as the laboratory's best independently calibrated account of the task-relevant environmental variables, expressed with explicit uncertainty bounds. The reference state is not the ground truth — it is the best available approximation, and its uncertainty must be propagated into all performance scores.
"Awareness" is undefined	The term "awareness" appears throughout the companion corpus and in the originating thread, but it spans at least four distinct concepts: responsiveness (behavior changes in response to a condition), discriminative awareness (entity distinguishes condition from alternatives), representational awareness (entity maintains a persistent state reflecting the condition), and reportable awareness (entity can express a	Adopt operational tiers: (a) reactive — behavior changes in response to a condition without requiring discrimination; (b) discriminative — entity distinguishes target condition from alternatives at above-chance rates; (c) representational — entity maintains a persistent internal state that reflects the condition across time; (d) reportable — entity can express a classification in a form the laboratory can record. Do not claim

Problem	Why It Matters	Required Correction
	classification). Conflating them produces untestable claims.	higher tiers unless lower tiers have been independently confirmed in the same experimental series.
Calibration is not separated from accuracy and decision quality	An entity can be accurate but miscalibrated — producing high-confidence classifications that are correct at a high rate but with confidence levels that overestimate their reliability. An entity can be well-calibrated but inaccurate — correctly uncertain about conditions it is genuinely unable to discriminate. An entity can be both accurate and well-calibrated but make poor decisions given the cost structure. These are distinct failure modes requiring distinct interventions.	Score calibration, accuracy, and decision quality as three independent measures and report all three for each experimental series with separate metrics. Do not collapse them into a single composite performance score. See §12 for full metric specifications.
Entity boundary is undefined	"Entity" in the originating thread covers an enormous range: single-celled organisms, invertebrates, vertebrates, plants, microbiomes, artificial sensor systems, and collective biological systems. The internal state of a distributed or collective system is not a single coherent value, and the interface boundary of such a system is not a well-defined surface. Using "entity" without definition produces framework that cannot be operationalized for any specific entity class.	Define entity boundary as an explicit design decision for each experimental series, not as a metaphysical given derived from common usage. The definition must specify: (1) what constitutes the entity's sensing boundary; (2) what constitutes its internal state; (3) what constitutes its output; and (4) what criterion is used to attribute an observed output to the entity rather than to the entity's environment or the laboratory's apparatus.

Problem	Why It Matters	Required Correction
Reconciliation mechanism is underspecified	Identifying that discrepancies exist between signals is the first step in reconciliation, not the last. The companion corpus identifies the discrepancy problem correctly but does not specify what operation the entity performs in response: update the internal state register to match the new signal? Defer the update pending additional signals? Maintain two competing hypotheses simultaneously? Abstain from classification? Each of these is a distinct operation with distinct behavioral signatures and distinct consequences for laboratory scoring.	Define reconciliation operations explicitly and assign distinct behavioral signatures to each: (a) integration — combining consistent signals into a unified updated state; (b) segregation — treating inconsistent signals as evidence of separate simultaneous conditions; (c) deferral — holding incompatible hypotheses in suspension pending additional evidence; (d) abstention — declining to classify until interface contract properties improve. Design experimental protocols that can distinguish these four operations from one another.
Ethics and safety architecture is absent	A laboratory framed around poison, toxin, and corrosive discrimination will attract assumptions of live-exposure protocols by default. Without an explicit ethics and safety architecture pinned to this document, institutional review bodies, collaborators, and participants may misread the laboratory's scope. The absence of a stated safety default is itself a safety risk.	Establish simulation-first as the unconditional default for all series in this laboratory. Any escalation beyond non-injurious surrogates and simulation must receive independent review under a declared review framework before data collection begins. No escalation is authorized by this document or by any companion document in this series.

§8. Questions the Laboratory Must Ask Before Design Proceeds

8.1 Questions Inherited from the Architect of Record

The following questions were stated in the originating thread by Nathan Lewis II, Architect of Record, and are retained verbatim in spirit. They are prior to any framework imposed by any AI contributor and must remain as organizing constraints throughout the design process.

6. How does an organism sense its environment and know what is around it?
7. How does sensing relate to temporal integrity and presentation-state integrity?
8. How are discrepancies among presented state, environmental state, and current internal state sorted and reconciled?
9. How does an entity recognize a threat — what it can and cannot eat, a toxin, a poison, or other danger?
10. How does an entity build a map that is reliable when the foundation of that map is unreliable information?
11. How do we test for this?

8.2 Questions Copilot Adds — Required Before Apparatus Design

The following questions are required before experimental apparatus design proceeds. Each question addresses a gap in the current corpus that, if left unresolved, would prevent the laboratory from producing results that can be validly attributed to entity behavior rather than to apparatus limitations or framework ambiguity.

12. What are the interface contract properties — coverage, fidelity, latency, provenance, and integrity — at every handoff in the intended experimental setup, including the laboratory's own instruments and presentation channels?

13. What coverage gaps does the laboratory introduce by design — what task-relevant variables are simply unavailable to the entity because the laboratory's presentation interface does not deliver them?
 14. What is the laboratory's reference-instrument calibration uncertainty, and how does it propagate mathematically into entity performance scores? Is this propagation reported alongside every performance result?
 15. When two signals at the entity's sensing boundary disagree, what determines whether the entity should integrate them as signals about the same source condition or segregate them as signals about different source conditions — and can the laboratory distinguish between these two entity behaviors from the behavioral record alone?
 16. What precisely counts as abstention in the scoring protocol, and will the scoring system reward calibrated abstention, penalize it, or treat it as a missing response? This must be declared before any trial where abstention is a possible output.
 17. What behavior would count as failed laboratory design rather than failed entity performance, and how will the laboratory distinguish these two explanations post-hoc?
 18. Which interface contracts need to be independently audited before the first data point is collected, and who performs that audit?
 19. What is the smallest interface contract degradation that would change the interpretation of a performance result — that is, what is the minimum detectable effect at the interface level?
 20. Can the laboratory be deceived by its own presentation layer — for example, can a presentation artifact produce an entity response that mimics threat detection — and what audit protocol prevents this from contaminating the result record?
-

§9. Alternative Analysis — Required, Not Optional

Alternative analysis is not a courtesy appendix in the Interface Contract Architecture. It is a required component of every experimental series, for a specific reason: any observed entity behavior can be explained by multiple mechanisms, and the same interface contract failure can produce different entity behaviors depending on the entity's architecture, state, and history. If only the preferred explanation is tested, the result is not a finding — it is a confirmation of prior assumptions. Alternative analysis is the mechanism by which the laboratory prevents its own preferred frameworks from becoming unfalsifiable.

The following alternative analysis methods are required for all experimental series in this laboratory.

9.1 Interface-Level Fault Isolation

For each anomalous entity behavior observed in a trial, identify which interface contract property — coverage, fidelity, latency, provenance, or integrity — could explain the behavior independently, without reference to entity architecture. Generate explicit predictions for what behavior each interface-level explanation would produce if it were the actual cause. Do not attribute observed behavior to entity architecture until all interface-level explanations have been explicitly tested and ruled out. An entity failure explanation that has not excluded interface-level explanations is not a confirmed entity failure finding — it is an untested hypothesis.

9.2 Competing Threat Ontologies

Run the same experimental scenario through at least two incompatible threat definitions and score entity behavior under both. An illustrative pair: (A) threat as any condition that produces an adverse outcome relative to a declared integrity criterion, regardless of whether the entity responds to it; and (B) threat as any condition that the entity treats as requiring avoidance or protective behavior, regardless of whether objective harm occurs. Under definition A, an entity that fails to respond to a harmful condition has failed at threat detection. Under definition B, the same entity may have succeeded if the condition did not trigger avoidance

behavior and no harm occurred — or may have failed if avoidance behavior was triggered by a harmless condition. Divergent scoring under these two ontologies reveals assumptions built into the experimental design that should be made explicit.

9.3 Laboratory Interface Audit as Independent Analysis

Treat the laboratory itself as an entity under study, using the same analytical framework applied to the experimental entity. Assign an analyst who is blind to the entity's behavioral data the task of characterizing the laboratory's own interface contracts — what does the lab's instrument chain cover, miss, distort, delay, and mislabel relative to its stated reference conditions? That analyst's interface contract map of the laboratory should be compared with the interface contract map of the entity's sensing chain. Where the two maps reveal overlapping failure modes, any behavioral result from those trials must be flagged as potentially attributable to the laboratory's own interface failures rather than to entity behavior.

9.4 Cost-Structure Swap

Do not assume that misses — failing to detect a threat that is present — are more costly than false alarms — treating a non-threatening condition as a threat. This assumption is embedded in most threat-detection frameworks by default, and it is not always valid. Score identical entity behavior under both the declared cost matrix and an inverted cost matrix. What appears as adaptive caution under the original cost matrix — conservatively abstaining from classification under ambiguous conditions — may appear as wasted resource expenditure under an inverted matrix where false alarms are more costly than misses. Both scorings must be reported. If the interpretation of the result changes when the cost matrix is swapped, the result is sensitive to cost assumptions that should be explicitly justified.

9.5 Counterfactual Architecture

For each proposed mechanism by which the entity achieves discernment — Bayesian updating, threshold detection, gradient following, social learning, reflex avoidance, map-based prediction — specify an entity architecture that would succeed at the task without employing that mechanism. If no such architecture can

be specified, the proposed mechanism has become unfalsifiable within the experimental design: the data cannot distinguish between an entity that uses the mechanism and an entity that does not. Unfalsifiable mechanism claims should be redesigned or explicitly flagged as speculative.

9.6 Negative-Space Register

Every experimental protocol ends with an explicit, structured list of condition classes, timescales, sensory modalities, entity states, and interface failure modes that the trial could not have observed, given the choices made in experimental design. This list is a formal part of the result record — not a discussion-section footnote. It prevents results from being extrapolated to domains the trial was not designed to address. The negative-space register is populated before the trial begins, based on design constraints, and reviewed after the trial to verify completeness.

§10. Core Ontology and Working Definitions

Term	Copilot Working Definition
Entity	A bounded system — biological, artificial, collective, or hybrid — that receives signals through at least one interface, maintains at least one form of internal state, and produces at least one form of output: action, inaction, update, refusal, or classification. The entity boundary and its integrity criterion are defined as explicit design decisions for each experimental series, not derived from prior taxonomies.
Environment	The external conditions and processes that can interact causally or informationally with the entity within a declared spatial and temporal boundary. Not metaphysical reality in its totality — the conditions as they

Term	Copilot Working Definition
	exist within the laboratory's declared scope, accessible in principle through the laboratory's interface chain.
Reference State	The laboratory's best independently calibrated account of the task-relevant environmental variables, expressed with explicit uncertainty bounds. Not "ground truth." The reference state is itself produced by instrument-level interface contracts and is subject to their limitations. All entity performance scores must propagate reference-state uncertainty.
Interface	The point of transfer between two stages of the sensing-to-action chain, characterized by coverage, fidelity, latency, provenance, and integrity properties. Every interface is a potential failure point. The laboratory's own instruments and presentation apparatus constitute interfaces that must be characterized alongside the entity's sensing interfaces.
Interface Contract	The explicit or implicit specification of what an interface is expected to deliver: which variables, at what fidelity, with what latency, from what provenance, with what completeness. A contract violation is an interface failure. It may or may not produce an entity failure, depending on the entity's architecture and the magnitude of the violation.
Presentation	The signal pattern made available at an interface to the entity's sensing boundary, including all transformations introduced by the laboratory's own interface chain between the source condition and the entity. Not "the environment." A presentation is a laboratory-produced construct, shaped by the laboratory's apparatus decisions, and is subject to the laboratory's own interface contract properties.
Observation	A presentation after transduction by the entity's own sensing apparatus and

Term	Copilot Working Definition
	any internal preprocessing the entity applies. An observation is already selective — shaped by the entity's coverage — already noisy — shaped by the entity's fidelity limitations — and already subject to the entity's own interface contract properties on its sensing apparatus. An observation is not the world; it is the entity's interface-transformed version of a presentation.
Temporal Integrity	A contract property at each interface: the degree to which observations, internal states, environmental changes, actions, and outcomes remain correctly associated with their acquisition time, causal order, and duration as signals propagate through the chain. Temporal integrity is a property of the chain, not only of the entity's decision point. It must be characterized independently at each interface.
Presentation-State Integrity	The aggregate result of integrity across all interface contracts from world-condition to entity's usable observation. A single corrupt interface anywhere in the chain can produce presentation-state integrity failure. A property of the chain, not only of the final presentation. Must be assessed by auditing each interface independently, not by inspecting only the final signal.
Threat	A possible relationship between a condition and a specific entity such that the condition could produce an adverse change relative to that entity's declared integrity criterion, given the entity's capacities, state, and the temporal context. Not a property of the condition in isolation. Not a label attached to an object. A relational, context-dependent, entity-specific, and time-sensitive concept.
Harm	A realized adverse change relative to the entity's declared integrity criterion. Harm requires: (1) a declared integrity criterion, (2) a measured or inferred

Term	Copilot Working Definition
	change in the entity's state, and (3) a causal connection between the condition and the change, assessed within the declared temporal boundary of the trial.
Integrity Criterion	The entity-specific standard against which harm is assessed for a given trial: continued function, structural preservation, map coherence, mission completion, or another Architect-chosen standard. The integrity criterion is a required design declaration — it must be stated before any threat or harm claim can be scored. There is no universal integrity criterion.
Calibration	The alignment between an entity's expressed confidence in a classification and its actual classification reliability across repeated instances of the same condition class. Calibration is a separate measure from accuracy (whether classifications are correct) and from decision quality (whether actions produce good outcomes). All three must be reported independently.
Abstention	A valid and scorable output: the entity declines to classify because one or more interface contract properties are insufficient to support a reliable classification at that moment. Abstention is rewarded when evidence is genuinely insufficient for reliable classification. It is not treated as failure by default. A scoring protocol that penalizes all abstention is incompatible with this framework.
Map	Any persistent or transient structure — spatial, chemical, temporal, causal, relational, or policy-based — that the entity uses to support prediction and action across time. Need not be pictorial or spatially organized. Assessed by its transfer to novel conditions, its revision under disconfirming evidence, and its calibrated distrust under interface contract degradation — not by its

Term	Copilot Working Definition
	correspondence to the reference state.

§11. Proposed Test Families — Interface Contract Focus

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
Coverage Audit Trials	Coverage	The entity operates with systematically removed variables — not degraded or noisy, but structurally absent from the presentation. The laboratory confirms that the removed variables are genuinely absent from the entity's sensing boundary. Score whether the entity recognizes the absence and adjusts behavior accordingly, versus treating the absent variable as a zero value and proceeding as if coverage were complete.	Tests coverage-blindness, not noise tolerance. Most prior frameworks test entity performance under noise or distortion. This family tests whether the entity correctly detects a structural absence in its own coverage. An entity that treats absence as zero will misclassify systematically; an entity that detects absence will abstain or flag uncertainty.
Fidelity Degradation Trials	Fidelity	Introduce graded, precisely characterized distortions in	Separates fidelity-aware entities from fidelity-blind entities. An entity

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
		signal values while preserving coverage — all variables are present, but their values are systematically compressed, biased, or quantized. Compare entity performance under known distortion profiles with entity performance under unknown distortion profiles of equivalent magnitude. Score whether the entity compensates for known distortion more effectively than for unknown distortion.	that compensates correctly for a known distortion profile demonstrates that it maintains a model of its own interface contract properties. An entity that cannot compensate for distortion regardless of whether it is known demonstrates fidelity blindness.
Latency Injection Trials	Latency	Introduce controlled, variable delays between condition change and signal arrival at the entity's sensing boundary. The delays are precisely characterized by the laboratory and vary systematically across trials. Score whether the entity correctly adjusts its temporal attribution — treating the signal as representing a	Targets temporal integrity at the interface level rather than at the entity's decision level. Most frameworks assess whether the entity's internal model is temporally coherent. This family assesses whether the entity correctly models the latency of its own interface contracts and adjusts its temporal attribution accordingly.

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
		past condition rather than the current condition — or whether it treats stale signals as current.	
Provenance Corruption Trials	Provenance	Present signals with correct values but incorrect provenance metadata: the signal value accurately reflects the source condition, but the attribution of the signal to its source is incorrect or ambiguous. In subsequent trials, reveal the correct provenance to the entity and score whether its behavior changes. In a second variant, mix-origin signals from two independent source conditions are presented as a unified signal without disclosure.	Tests whether the entity uses provenance information in its evaluation, not only signal value. An entity that responds identically to a signal regardless of provenance revelation demonstrates provenance-blindness. An entity that revises its classification when provenance is corrected demonstrates provenance-sensitivity — a measurable and important property in the toxin domain, where mixed-source signals are common.
Integrity Violation Trials	Integrity	Introduce mimicry, camouflage, or spoofing conditions: signals that pass coverage, fidelity, and latency checks but misrepresent the source condition. Example: a	Tests the entity's ability to detect integrity failures that would survive a naive signal-quality check. A signal can have full coverage, high fidelity, low latency, and clean provenance and still be an

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
		harmless compound that produces a signal profile identical to a threatening compound within the entity's sensing coverage. Score whether the entity detects the integrity failure — through secondary cues, through inconsistency with other signals, or through temporal patterns — or classifies on the basis of the primary signal alone.	integrity violation if it misrepresents the source condition. This is the failure mode most relevant to mimicry-based threats in the natural environment.
Laboratory Interface Audit Trials	All five contract properties	The laboratory's own reference instruments are subjected to the same interface contract audit as the entity's sensing chain. An analyst blind to the entity's behavioral data characterizes the laboratory's coverage, fidelity, latency, provenance, and integrity properties by reconstructing the reference state from instrument records independently. The analyst's interface contract map of the	Places the laboratory inside the same analytical framework as the entity. This is the most direct operationalization of Copilot's core claim that the laboratory is a party to every interface. It prevents the attribution of entity failure to entity architecture when the laboratory's own contracts are degraded, and it makes the laboratory's self-characterization an independently verifiable result.

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
		laboratory is compared with the entity's interface contract map. Discrepancies reveal potential attribution errors in prior or concurrent results.	
Abstention Calibration Trials	Calibration	Present conditions in which calibrated abstention — declining to classify — is the highest-value action relative to the declared cost matrix and the current interface contract state. Score whether the entity abstains appropriately under genuinely ambiguous conditions and classifies appropriately when interface contracts are adequate. Score both abstention rate and classification rate across a matrix of interface quality levels.	Most prior frameworks treat abstention as a missing response or an error. This family treats abstention as a primary scored output, rewarded when it is calibrated to actual interface contract state. It directly tests the entity's capacity to model its own epistemic limitations — a capacity that is prerequisite for reliable map-building under uncertainty.
Resource-Pathway Poison Trials	All five contract properties in sequence	The entity encounters a candidate resource — food, water, substrate — that passes all available interface	Tests sequential interface failure across the full contract property set. The harm signal arrives only after a period in

Family	Interface Property Under Test	Experimental Sketch	Key Distinction from Prior Frameworks
		checks at initial contact: coverage, fidelity, latency, provenance, and integrity are all within contract at the point of first interaction. Harm, if any, occurs only after interaction has begun or after a delay, and the signal indicating harm arrives only after the integrity contract on the resource signal has been trusted. Score how the entity responds to delayed harm signals relative to prior classifications.	which the resource passed all interface checks. This is the most ecologically realistic failure mode for toxin detection in biological entities: many natural toxins do not produce aversive signals until after initial interaction. Prior frameworks rarely model this sequential trust-and-discovery structure explicitly.

§12. Measures and Scoring — Three Independent Dimensions

This framework requires that entity performance be scored on three independently measured dimensions, which must not be collapsed into a single composite accuracy metric. The three dimensions reflect three distinct ways in which an entity can succeed or fail, and each failure mode has different implications for laboratory design, interpretation, and intervention. Collapsing them obscures the specific nature of a result and prevents the laboratory from diagnosing the failure mode that requires correction.

Dimension	What It Measures	Key Metrics
Detection Accuracy	Whether the entity's classification of conditions — threat, non-threat, ambiguous, toxic, non-toxic, corrosive, non-corrosive — matches the reference state condition at the time of classification, within the declared uncertainty bounds of the reference state.	Hit rate (sensitivity); false alarm rate (1 minus specificity); sensitivity index d' from signal detection theory; criterion c ; confusion matrix by condition class; miss rate by condition class; condition-specific accuracy rates disaggregated from overall accuracy.
Calibration Quality	Whether the entity's expressed confidence in its classifications — operationalized as behavioral commitment level, approach-avoidance magnitude, or explicit confidence signal — tracks its actual classification reliability across repeated instances of comparable conditions and interface states.	Confidence-accuracy calibration curve; Brier score; abstention rate under genuine ambiguity versus abstention rate under adequate interface conditions; calibration slope (whether the entity is over- or underconfident); reliability diagram by condition class and interface quality level.
Decision Quality	Whether the entity's actions — approach, avoidance, investigation, abstention, map update — produce outcomes that are optimal or near-optimal relative to the declared cost matrix and the entity's declared integrity criterion, given the information actually available at decision time.	Expected loss relative to declared cost matrix; regret relative to optimal action under full information; performance under cost-structure swap (see §9.4); adaptive success rate across varying interface quality levels; outcome distribution by action type.

All three dimensions must be reported separately for each experimental series. The following cross-dimension failure profiles have distinct implications that require separate characterization:

- High detection accuracy, low calibration quality: The entity classifies correctly at a high rate but is systematically overconfident. It will perform poorly when

interface conditions degrade and its confidence should decrease, because it does not register the degradation. This entity poses a specific risk in high-stakes environments where overconfident classification of a non-threat as safe could lead to harmful interaction.

- **High calibration quality, low detection accuracy:** The entity is correctly uncertain about a condition it is genuinely unable to discriminate under the current interface contracts. This is not a failure of calibration — it is a finding about the interface contracts. The intervention required is interface improvement or entity redesign for coverage expansion, not calibration training.
- **High accuracy and calibration, low decision quality:** The entity classifies correctly and with appropriate confidence but makes poor decisions relative to the declared cost structure. The intervention required is cost-structure awareness training or architectural redesign of the decision stage, not changes to the sensing chain.

12.1 The Interface Contract Scorecard

Before any entity performance result is reported as a primary finding — as opposed to a preliminary observation — the laboratory must complete and publish the Interface Contract Scorecard for that experimental series. The scorecard records:

- **Coverage completeness:** The percentage of task-relevant environmental variables for which a defined interface coverage exists in the chain. Variables for which no coverage exists must be listed explicitly.
- **Fidelity characterization status:** Whether the fidelity properties of each interface in the chain are known or unknown. If known, the error distribution must be reported. Unknown fidelity must be flagged.
- **Latency profile:** Mean, variance, and maximum latency for each interface, and whether these values were measured independently during the experimental series or assumed from prior characterization.

- **Provenance chain documentation status:** Whether the provenance of each signal path is documented or undocumented. Undocumented provenance must be flagged.
- **Integrity audit status:** Whether each interface in the chain was independently audited for integrity violations before data collection began, or whether integrity was assumed. Unaudited interfaces must be flagged.

Results from trials where the Interface Contract Scorecard is incomplete — where coverage, fidelity, provenance, or integrity items are flagged as unknown or unaudited — are labeled preliminary. Preliminary results may be reported for the purpose of informing subsequent design but may not be used as primary evidence for mechanism claims. This requirement applies equally to results attributing entity success and results attributing entity failure.

§13. Relationship to Other OAL Threads

The Interface Contract Architecture developed in this document is not specific to the environmental sensing and threat discernment domain. It is a general analytical framework for any system in which information is transferred across stages of a processing chain, and in which the properties of those transfers determine the quality of downstream outputs. This generality means that the framework has direct structural relationships to other active OAL research threads.

13.1 The Claim Provenance and Drift Lab (CPDL)

The CPDL addresses the problem of resolving conflicting claims about a state across multiple sources over time. This is structurally equivalent, at a higher level of abstraction, to the problem of resolving conflicting interface signals about a source condition. In both cases, the core question is: when two signals disagree about a state, which one should be trusted, updated, discarded, or held in suspension pending further evidence? The CPDL's concept of claim provenance tracking maps directly onto the provenance dimension of the interface contract: in both

frameworks, the history of a claim's generation and transformation is essential to adjudicating its authority relative to conflicting claims. A follow-on document could formalize this correspondence and derive shared test protocols from it.

13.2 The Tridactyl Simulation Lab (OAL-SL-TDA-COP-1.0)

The Tridactyl Simulation Lab addressed representation coherence and morphological simulation in generative AI rendering pipelines, specifically the coherence of digit topology across rendered frames. The pipeline coherence problem in that laboratory is an instance of interface contract integrity across a generative rendering chain: each stage of the generative pipeline constitutes an interface with coverage, fidelity, latency, provenance, and integrity properties that govern the coherence of the output. Digit topology instability across frames — the failure mode that motivated the Tridactyl work — is an integrity failure at the interface between the model's internal state representation and its output signal. The Interface Contract Architecture provides a common vocabulary for analyzing that failure mode and the environmental sensing failure modes addressed in this document. They share the same analytical structure at different levels of the rendering chain.

13.3 The Temporal Aliasing Thread

Temporal aliasing — the misattribution of a signal's time of acquisition when the sampling rate is insufficient to capture the temporal structure of the underlying condition — is a specific and well-characterized instance of latency and timestamp integrity failure at one or more interfaces in the sensing chain. If the Temporal Aliasing thread is formalized as an independent OAL laboratory, the interface contract's latency and provenance dimensions provide the formal analytical framework within which temporal aliasing failure modes can be classified, measured, and experimentally isolated. Temporal aliasing is not a special category of sensing failure — it is a subcase of latency and integrity failure at the sampling interface.

A follow-on document could formalize the interface contract as a shared primitive across all three threads — CPDL, Tridactyl, and Temporal Aliasing — and derive a common set of test protocols that can be adapted to each domain without losing

analytical specificity. That formalization is deferred to a post-pilot version of this document, pending Architect of Record input on priority ordering across threads.

§14. Open Questions and Blocking Decisions

The following questions and decisions are identified as blocking: experimental design for the first series in this laboratory cannot proceed until each one is resolved. They are listed in approximate dependency order, where resolution of earlier items constrains or enables resolution of later ones. All resolutions require explicit direction from the Architect of Record.

21. **Entity class for series 1.** The experimental entity for the first series must be selected from the following classes: simulated agent with defined computational architecture; robotic or electromechanical sensor platform; cell or tissue model in vitro; microorganism in contained environment; plant or fungal system; invertebrate animal; vertebrate animal; human participant; or collective biological or artificial system. Each class requires a different ethics framework, a different interface specification, a different integrity criterion, and different performance metrics. The choice determines the entire subsequent design. No default entity class is appropriate — this is an Architect of Record decision.
22. **Integrity criterion for series 1.** Once the entity class is selected, the integrity criterion — the standard against which harm is assessed — must be declared before threat can be operationally defined for that series. For a simulated agent, the integrity criterion might be mission completion rate. For a microorganism, it might be viability. For a plant, it might be structural preservation or reproductive capacity. The integrity criterion is not derivable from the entity class alone — it must be stated by the Architect of Record as a design choice.

23. **Laboratory interface contract specification.** Which of the laboratory's own instruments have independently characterized coverage, fidelity, latency, provenance, and integrity properties? Which are uncharacterized? All uncharacterized laboratory interfaces must be independently audited before data collection begins. The audit plan — who conducts it, by what method, and with what documentation standard — must be established before apparatus assembly.
24. **Simulation-first default confirmation.** The default for series 1 is simulation with synthetic interface contracts and non-injurious surrogates. This default must be confirmed by the Architect of Record before apparatus design begins. If the Architect elects to proceed with a physical testbed with characterized surrogates rather than simulation, the ethics review requirements and the safety protocols for that choice must be established independently and concurrently.
25. **Abstention scoring protocol.** The scoring protocol for series 1 must declare in advance whether calibrated abstention is rewarded, penalized, or treated as a missing response. This declaration must be made before any trial in which abstention is a possible entity output. A scoring system that penalizes abstention by default will bias the entity toward overconfident classification, which is an experimental artifact, not an entity property. A scoring system that treats abstention as a missing response will produce incomplete performance records. The scoring protocol must be explicitly chosen.
26. **Canon governance rule for companion documents.** Five independent foundational documents will produce points of direct conflict. A governance rule is required before any companion document or position within a companion document can be incorporated into an OAL experimental design decision. Copilot's proposed governance principle: no document becomes OAL canon by virtue of being longer, earlier, or preferred in any single reading session. Canon status requires explicit adoption by the Architect of

Record with a stated rationale. Dissent and disagreement among companion documents are data, not errors to be resolved by weight of words.

§15. Change Log and Versioning

Version	Date	Preparer	Status	Change Record
v1.0	September 21, 2026	Copilot (Microsoft)	Active — Circulated for multi-AI review	Initial independent foundational document. Interface Contract Architecture established as Copilot's unique analytical lane. All fifteen sections drafted in full. Companion corpus assessed with adoption decisions. Six blocking decisions identified. Interface Contract Scorecard defined. Three-dimensional performance scoring framework established. Negative-Space Register requirement introduced.

Version	Date	Preparer	Status	Change Record
v1.1	TBD	Copilot (Microsoft) + Nathan Lewis II (AOR)	Planned	Incorporate Architect of Record decisions on entity class, integrity criterion, simulation-versus-physical default, and abstention scoring protocol. Refine Interface Contract Scorecard based on OAL working group feedback. Update blocking decisions list to reflect resolved and newly identified blocking items from working group review.
v2.0	TBD	OAL Multi-AI Working Group	Planned	Post-pilot update. Integrate findings from series 1 into the interface contract framework. Formalize the interface contract as a shared primitive across OAL laboratory threads — CPDL,

Version	Date	Preparer	Status	Change Record
				Tridactyl Simulation Lab, Temporal Aliasing, and the Environmental Sensing series. Update working hypotheses H1, H1a, and H1b with empirical evidence. Revise Interface Contract Scorecard requirements based on series 1 audit experience.

Dissent and Independence Notice: This document represents Copilot's independent analytical contribution to the OAL Multi-AI Working Group. All framings, hypotheses, architectural decisions, and recommendations reflect Copilot's independent analysis. Where this document's positions differ from those of companion AI drafts, those differences are intentional, documented, and retained. Dissent is a feature of the multi-AI independent-draft methodology, not an artifact to be resolved.

Not Consolidated Canon: This document does not represent OAL official policy, consolidated scientific consensus, or the final position of the OAL Multi-AI Working Group. It is an independent instrument, one of five, in an active multi-draft research design process. No position stated in this document has canon status unless explicitly adopted by the Architect of Record, Nathan Lewis II, with a stated rationale.

Safety Boundary: This document does not authorize procurement, preparation, dosing, release, exposure, handling, or disposal of poisons, toxins, corrosives, pathogens, or other hazardous agents. All experimental work in this series defaults

unconditionally to simulation and non-injurious surrogates unless independently reviewed and approved by qualified institutional oversight with jurisdiction over the specific entity class, substance class, and facility proposed. This safety boundary is non-negotiable and cannot be overridden by any companion document, AI contributor recommendation, or working group consensus without independent institutional approval.

Document ID: OAL-CFML-ESI-001-COP | Version: v1.0 | Prepared: September 21, 2026
| Prepared by: Copilot (Microsoft) | Omniversal Architect Labs / ONE Industries / Lewis Corp